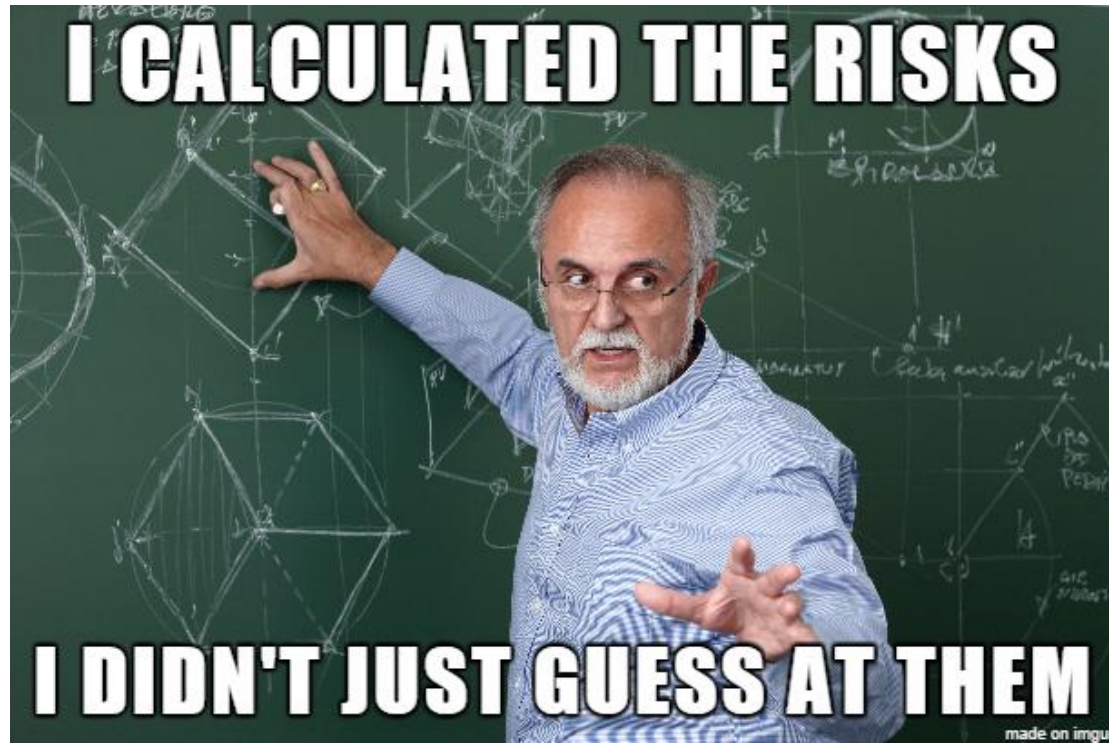


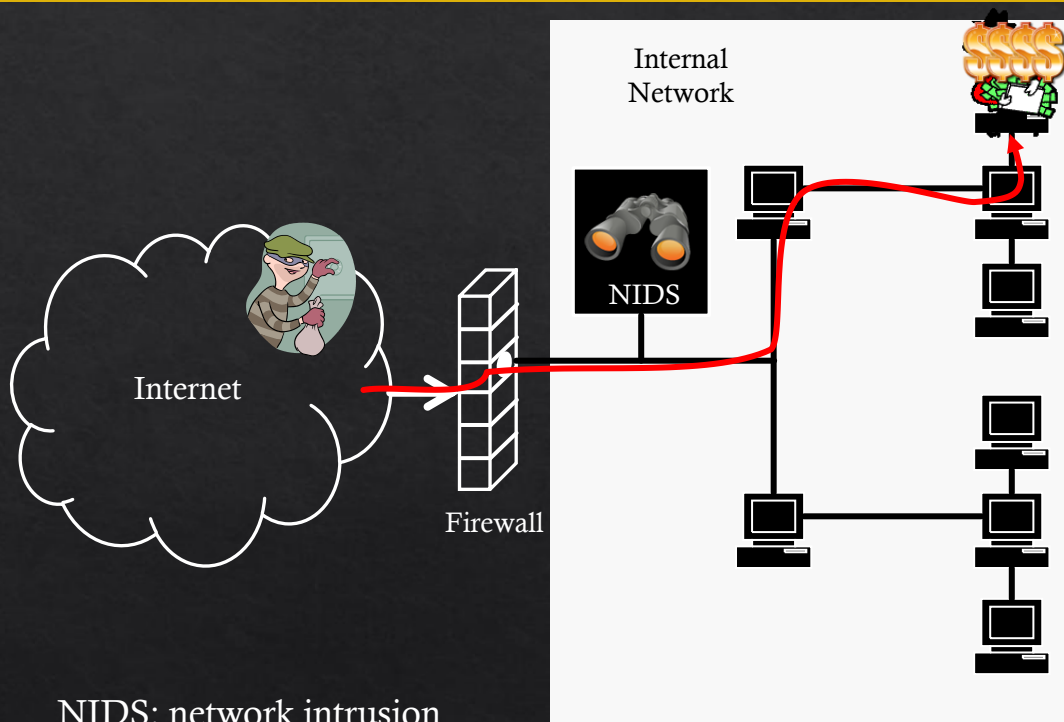


8. Threat Modelling

Garrison Gao

garrison.gao@uwa.edu.au

How secure is my system/network?



NIDS: network intrusion
detection system



Attacker-centric, Architecture-centric, Asset-centric

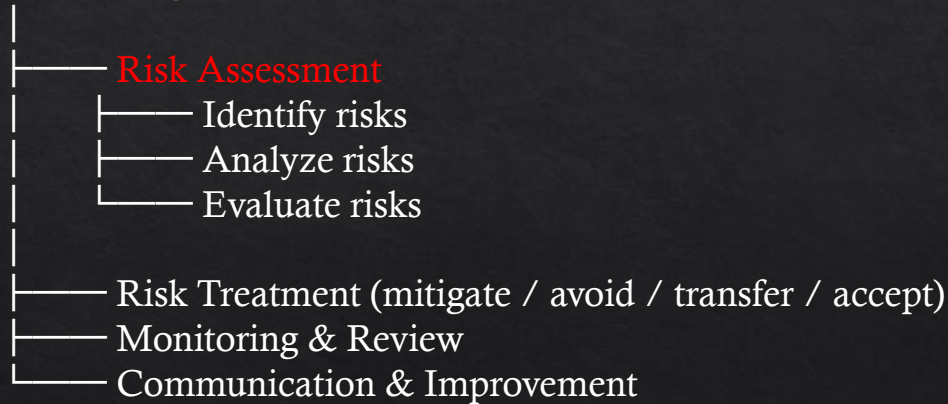
Threat Modelling Definition & Purpose

- ❖ **Threat modelling** is a process by which potential **threats**, such as structural vulnerabilities or the absence of appropriate safeguards, can be **identified** and **enumerated**, and **countermeasures prioritized**
- ❖ **The purpose** of threat modelling is to provide defenders with a systematic analysis of what controls or defences need to be included, given the **nature of the system**, the **probable attacker's profile**, the **most likely attack vectors**, and the **assets most desired** by an attacker.

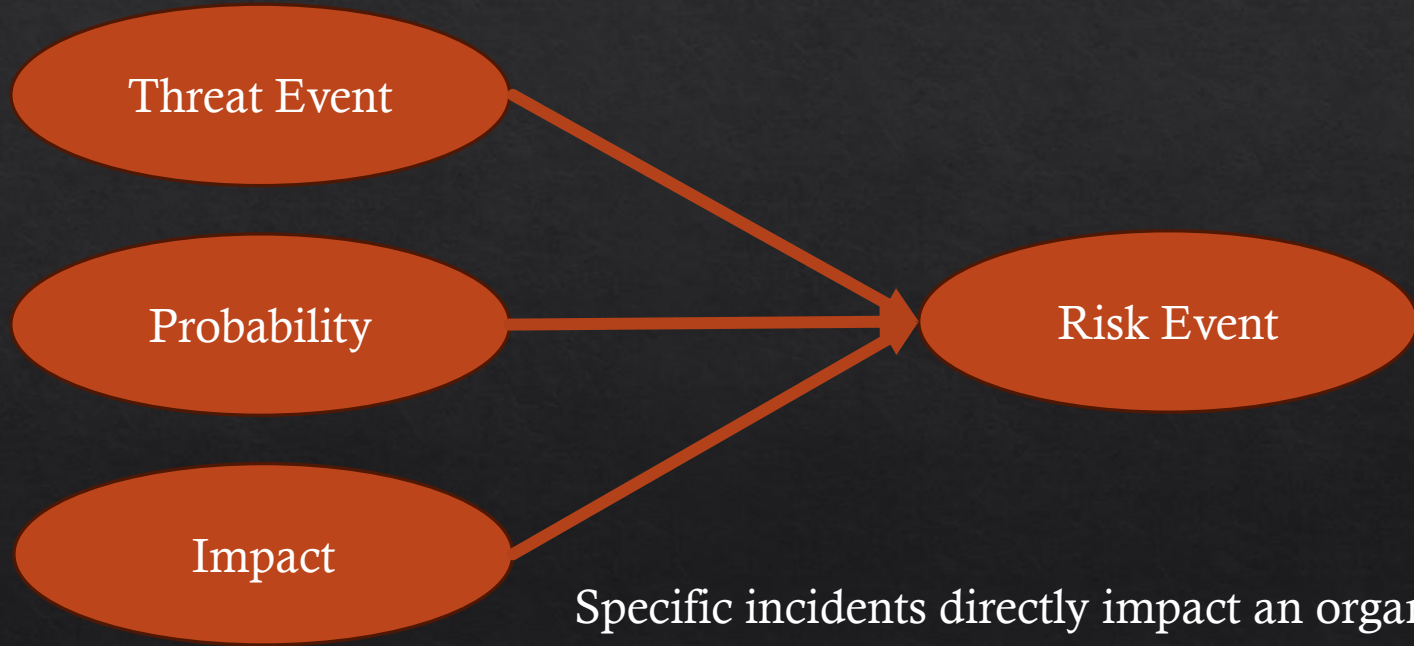
Risk Management Vs Risk Assessment

- ◇ **Risk Assessment:** it is a **specific step** within **risk management** that focuses on identifying risks, analyzing their likelihood and impact, and evaluating their severity.

Risk Management



Risk Events Vs Threat Event



Threat Modelling Vs Risk Assessment Vs Risk Management



- ◇ Threat Modeling → “What can go wrong?”
- ◇ Risk Assessment → “How bad and how likely is it?”
- ◇ Risk Management → “What should we do about it?”

Threat Modelling is Everywhere

Traveling to work

- ◆ Threat Modelling Steps:

- ◆ Assets:

- ◆ Threats:

- ◆ Mitigations (preemptive actions):

- ◆ -

- ◆ -

- ◆ -

- ◆ -

Threat Modelling is Everywhere

Traveling to work

◆ Threat Modelling Steps:

- ◆ **Assets:** Your safety, time, vehicle
- ◆ **Threats:** Traffic accidents, road congestion, bad weather, reckless drivers, leaving late, unfamiliar routes, poor vehicle condition
- ◆ **Mitigations (preemptive actions):**
 - ◆ Check traffic app before leaving
 - ◆ Leave earlier
 - ◆ Choose safer/less congested route
 - ◆ Drive cautiously in bad weather

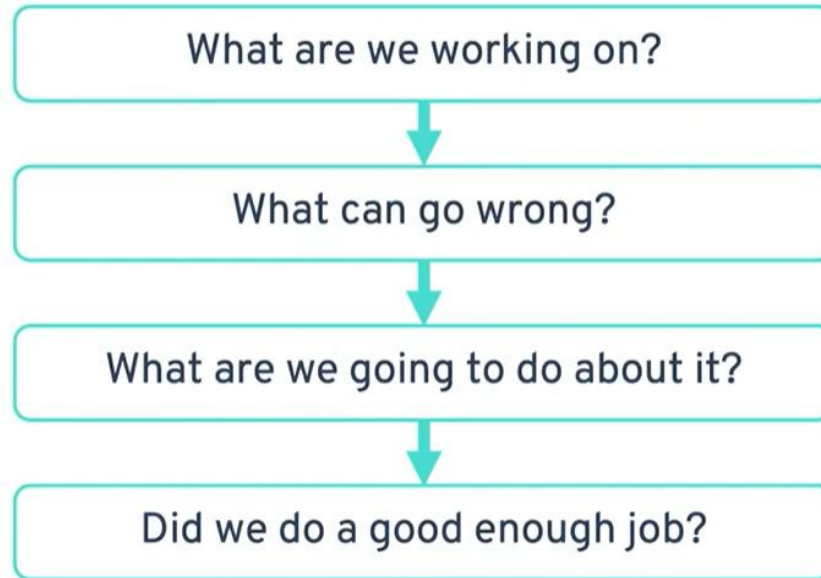
Threat Modelling is Everywhere

Preparing for an exam

- ◆ **Assets:** Grades, academic performance
- ◆ **Threats:** Forgetting material, running out of time, poor preparation, lack of practice
- ◆ **Mitigations:**
 - ◆ Practice past papers
 - ◆ Time management strategy
 - ◆ Review key concepts

Threat Modelling

Four key questions



Threat Modelling

What are we working on?

- ◇ For example, you might choose to model:
 - ◇ Software or applications
 - ◇ Enterprise architecture
 - ◇ Operational technology
 - ◇ Mobile apps
- ◇ What's important is **creating data- and process-flow diagrams** that help you share what's in your head with the people around you.



<https://threatmodelingbook.com/resources/threat-modeling>

Threat Modelling

What can go wrong?

- ◆ Some considerations include:
 - ◆ Who's the intended user?
 - ◆ What's the intended use?
 - ◆ Can an unintended user get access to it?
 - ◆ Can someone use it in an unintended way?
- ◆ The question “what can go wrong” helps you figure out the threats.

Threat Modelling

What are we going to do?

- ◆ Some types of actions include:
 - ◆ **Mitigating threats:** Make it harder for someone to take advantage of a threat.
 - ◆ **Eliminating threats:** Remove the feature or interface that creates the threat.
 - ◆ **Transferring threats:** Have someone else be responsible, like having a customer change default settings.
 - ◆ **Accepting threats:** Recognize that the time and effort to mitigate or eliminate the threat undermines the purpose of the project.

Threat Modelling

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 - ◆ **Accepting threats:** Recognize that the time and effort to mitigate or eliminate the threat undermines the purpose of the project.
- ◆ While these *sound* like, and are aligned to, risk management strategies, they are subtly different. Remember: threats are the things that *can* go wrong. Risk management focuses on the likelihood and impact of threats. You can use implicit agreement when choosing because it can be faster and easier than formal risk management approaches.

Threat Modelling

Did we do a good job?

- ◆ This step includes checking:
 - ◆ **The model**: make sure it matches what you built, starting with the diagram.
 - ◆ **Each threat**: review that you found all the possible threats and did the right thing with them.
 - ◆ **Tests**: ensure you have a good test to detect the problem, one that is in line with other software tests and the risks that failures expose.
- ◆ It validates the threat model by checking your work to make sure it's as complete as possible

Threat Modelling Techniques and Methods

- ◇ What we're working on
- ◇ How we're working on it

Kill Chain

Although the kill chain is often taught in incident response, it is actually very powerful proactively for threat modeling to answer “what can go wrong?”.

Those steps include:

- ◊ Reconnaissance – attacker gathers information
- ◊ Weaponization – prepares exploit/payload
- ◊ Delivery – sends the payload
- ◊ Exploitation – triggers vulnerability
- ◊ Installation – establishes persistence
- ◊ Command & Control (C2) – remote control channel
- ◊ Actions on Objectives – achieve goal (e.g., data exfiltration)

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What can go wrong (web application)

- ◊ Attacker scans open ports; Scrapes API endpointsRisk to authenticity
- ◊ Attacker crafts SQL injection payload
- ◊ Malicious request sent via web form/API
- ◊ SQL injection succeeds (lack input validation)
- ◊ Attacker uploads web shell
- ◊ Server connects to attacker infrastructure (outbound traffic is not checked)
- ◊ Data exfiltration, privilege abuse (sensitive data are not encrypted or segmented)

STRIDE

STRIDE is also a great way to answer, “what can go wrong.” With the STRIDE technique, you look at **different attack types** that software tends to experience.

STRIDE

◆ Spoofing

- ◆ Hijacking another user's identity
- ◆ Risk to **authenticity**
 - ◆ How could an attacker compromise a user's credentials?
 - ◆ What could an attacker do with those credentials?

<https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool-threats>
<https://www.linkedin.com/learning/static-application-security-testing/application-threat-modeling-stride?resume=false&u=2213369>

STRIDE

◊ Tampering

- ◊ Maliciously modifying data
- ◊ Risk to integrity
 - ◊ How could an attacker change stored data?
 - ◊ How could an attacker change data under transmission?

STRIDE

◇ Repudiation

- ◇ Denying that an action was performed
- ◇ Risk to **non-repudiation**
 - ◇ How could an attacker trigger transactions outside of your non-repudiation control?

STRIDE

◆ Repudiation

- ◆ Denying that an action was performed
- ◆ Risk to **non-repudiation**
 - ◆ How could an attacker trigger transactions outside of your non-repudiation control?

◆ Information Disclosure

- ◆ Exposing information to people who shouldn't see it
- ◆ Risk to **confidentiality**
 - ◆ How could an attacker read stored data?
 - ◆ How could an attacker read data under transmission?

STRIDE

◇ Denial of Service

- ◇ Denying access or services to authorized users
- ◇ Risk to **availability**
 - ◇ How could an attacker exploit application, infrastructure weaknesses to disrupt service availability?

STRIDE

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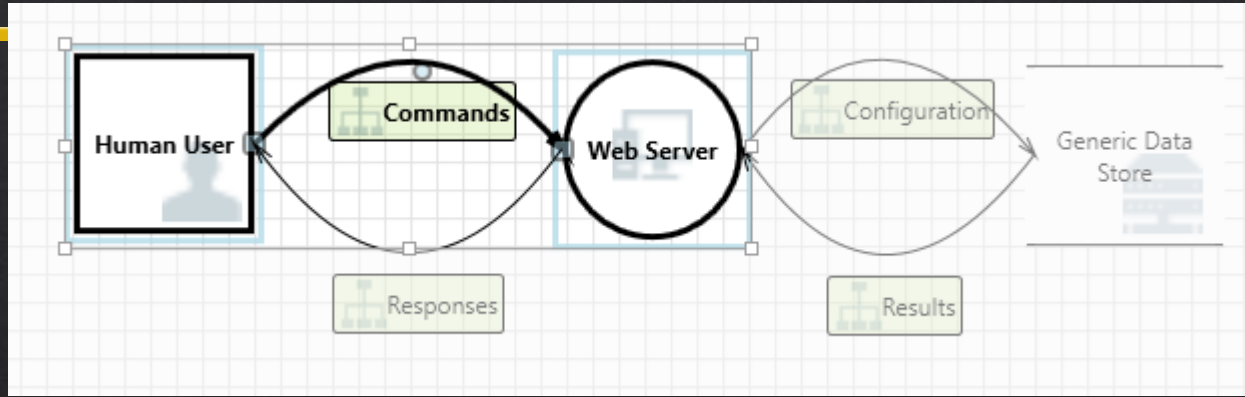
◆ Elevation of Privilege

- ◆ Abusing privileges
- ◆ Elevation = escalation
- ◆ Risk to **authorization**
 - ◆ How could an attacker compromise administrative accounts or interfaces?
 - ◆ How could an attacker run commands with elevated privileges?

Diagram

- ◆ Data Flow Diagram
- ◆ It is highly associated with threat modeling, in part because they're simple to use to show how data flows. They're a great way to show “**what we're working on.**”

Microsoft Threat Modeling Tool (DFD)



Threat Properties			
ID: 0	Diagram: Diagram 1	Status:	Not Started
Title:	Spoofing the Human User External Entity		
Category:	Spoofing		
Description:	Human User may be spoofed by an attacker and this may lead to unauthorized access to Web Server. Consider using a standard authentication mechanism to identify the external entity.		
Justification:			
Interaction:	Commands		
Priority:	High		

DREAD

- ❖ DREAD focuses on impact and risk, not identifying threats. In the past, people used DREAD to help organize and prioritize answers to the question, “what are we going to do about it?”
- ❖ We rank these factors on a scale from 0-10 and calculate the sum of these values. If the resulting value is higher, the risk of a potential attack on the organization is greater and we need to employ mitigation strategies immediately.
- ❖ This method lacks the ability to help you define factors for impact and likelihood that relate to a formal risk management approach.

<https://www.practical-devsecops.com/dread-threat-modeling/?srsltid=AfmBOopnPSLKYSg4XmQJazb8nhgJer-JLQKwvIGZiTYadhh1fNDyPdE>

DREAD

- ◆ Damage attribute

- ◆ Maps to **Impact**

- ◆ How big would the damage be if the attack succeeded?
 - ◆ Can an attacker completely take over and manipulate the system?
 - ◆ Can an attacker crash the system?

DREAD

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 - ◆ Can an attacker completely take over and manipulate the system?
 - ◆ Can an attacker crash the system?

- ◆ Reproducibility

- ◆ Maps to **Likelihood**

- ◆ How easy is it to reproduce an attack to work?
 - ◆ Can the exploit be automated?

DREAD

- ◆ Exploitability

- ◆ Effort Required

- ◆ How much time, effort, and expertise is needed to exploit the threat?
 - ◆ Does the attacker need to be authenticated?

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DREAD

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- ◆ Affected Users

- ◆ Impacted audience

- ◆ If a threat was exploited, what percentage of users would be affected?
 - ◆ Can an attacker gain administrative access to the system?

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- ◆ Discoverability

- ◆ Visibility

- ◆ How easy is it for an attacker to discover this threat?

How to Apply DREAD

- ◇ Score each threat
 - ◇ Prioritize remediation efforts quantitatively
 - ◇ Make decisions about how and when to fix based on scores

Security Cards

- ◇ Security Cards uses a deck of 42 cards to create threat scenarios based on four dimensions:
 - ◇ **Adversary Motivations:** reasons threat actors would attack a system
 - ◇ **Adversary Resources:** Skills, hardware, and software actors might use as part of an attack
 - ◇ **Adversary Methods:** How threat actors might approach an attack, including using technology and social engineering
 - ◇ **Human Impact:** What happens to people as the result of an attack
- ◇ Designed to **stimulate creative, human-centered threat thinking**, especially in early design stages.

<https://securitycards.cs.washington.edu/index.html>



How to Apply Security Cards

Typical usage:

- ◇ Draw 1–2 cards from each category
- ◇ Combine them into a scenario
 - ◇ Ask: Is this realistic?
 - ◇ What system weaknesses enable it?
- ◇ Derive:
 - ◇ Threats
 - ◇ Risks
 - ◇ Mitigations
- ◇ It's essentially guided brainstorming with structure

How to Apply Security Cards

System: Online Banking App:

- ◆ Draw cards:
 - ◆ Motivation: Financial gain
 - ◆ Resources: Low-skilled attacker with phishing kits
 - ◆ Methods: Social engineering (phishing email)
 - ◆ Human Impact: Loss of savings + emotional distress

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Step 2: Link to Threat Modeling

- ◆ Assets: User accounts, money
- ◆ Threat: Credential theft
- ◆ Vulnerability: Users trust phishing emails
- ◆ Attack path:
 - ◆ Delivery → phishing email
 - ◆ Exploitation → user enters credentials
 - ◆ Action → unauthorized transfer

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Step 3: Risk Assessment

- ◇ Likelihood: High (common attack)
- ◇ Impact: High (financial + psychological)
- ◇ Vulnerability: Users trust phishing emails

Step 4: Risk Management (Mitigation)

Security Cards

Security Cards transform threat modeling from a rigid analysis into a **creative, scenario-driven** exploration of adversaries, their capabilities, and real human consequences.

Method

STRIDE

Kill Chain

Security Cards

Strength

Systematic classification

Attack sequence

Creative scenario generation

Each method complements others

Persona non Grata (PnG)

- ◇ PnG is a lightweight, **attacker-centric** threat modeling technique. It's particularly useful for answering: "Can someone use this system in an unintended or abusive way?" Unlike structured methods (e.g., STRIDE), PnG starts from the human attacker perspective, not the system. It looks at attacker's:
 - ◇ Motivation
 - ◇ Skills
 - ◇ Goals
- ◇ The key shift:
 - ◇ Not "what vulnerabilities exist?"
 - ◇ But "how would this person misuse the system?"

How to Apply PnG

How PnG Works (Process)

- ◇ Define a **persona**
- ◇ Specify:
 - ◇ Motivation
 - ◇ Skills
 - ◇ Goals
- ◇ Ask:
 - ◇ “How would this person use the system in unintended ways?”
- ◇ Identify:
 - ◇ Misuse cases
 - ◇ Weak points
 - ◇ Missing controls

Example: University Learning Management System (LMS)

Step 1: Define Persona

- ◇ **Persona:** Disgruntled student
 - ◇ **Motivation:** Revenge after failing a course
 - ◇ **Skills:** Moderate (basic scripting, understands LMS workflow)
 - ◇ **Goals:**
 - ◇ Access exam answers
 - ◇ Disrupt grading process

How to Apply PnG

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Example: University Learning Management System (LMS)

Step 2: Misuse Thinking

- ◇ Ask:
 - ◇ “How could this student misuse the LMS?”
- ◇ Possible unintended uses:
 - ◇ Exploit weak access control to view hidden exam materials
 - ◇ Automate repeated requests to overload grading system
 - ◇ Share login credentials with others to bypass restrictions
 - ◇ Abuse discussion forums to spread misinformation

How to Apply PnG

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Example: University Learning Management System (LMS)

Step 3: Identify Vulnerabilities

- ◇ Poor role-based access control
- ◇ Predictable URLs for exam files
- ◇ Weak authentication policies

How to Apply PnG

How PnG Works (Process)

- ◇ Define a **persona**
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Example: University Learning Management System (LMS)

Step 4: Derive Risks

- ◇ Exam leakage → academic integrity loss
- ◇ System disruption → grading delays
- ◇ Reputation damage

Step 5: Mitigation (Risk Management)

- ◇ Enforce strict access control
- ◇ Use randomized resource identifiers
- ◇ Enable multi-factor authentication

CMU's Hybrid Threat Modeling Method (hTMM)

- ◇ The **CMU Hybrid Threat Modeling Method (hTMM)** is essentially a *pragmatic pipeline* that combines:
 - ◇ **Creativity** (Security Cards)
 - ◇ **Attacker realism** (PnG)
 - ◇ **Systematic structure** (STRIDE)
- ◇ It is used to answer the core question:
 - ◇ **“What can go wrong?”** → then refine → then formally assess risk

CMU's Hybrid Threat Modeling Method (hTMM)

Why hTMM Exists

- ◇ Each individual method has limitations:
 - ◇ **STRIDE** → systematic but can be rigid
 - ◇ **Security Cards** → creative but can be noisy/unfocused
 - ◇ **PnG** → realistic but depends on good personas
- ◇ hTMM combines them to:
 - ◇ Generate **many threat ideas**
 - ◇ Filter them into **realistic scenarios**
 - ◇ Structure them for **risk assessment**

The hTMM Process

Step 1: Identify the System

- ◇ Define:
 - ◇ Assets
 - ◇ Components
 - ◇ Users
 - ◇ Trust boundaries
- ◇ This anchors the analysis.

Step 2: Apply Security Cards (Divergent Thinking)

- ◇ Use cards to generate **broad threat scenarios**:
 - ◇ Motivation
 - ◇ Resources
 - ◇ Methods
 - ◇ Human impact
- ◇ At this stage:
 - ◇ **Quantity > quality**
 - ◇ Expect **many unrealistic ideas**

The hTMM Process

Step 3: Filter Using PnG (Realism Check)

- ◇ Now introduce **attacker personas**:

- ◇ Motivation

- ◇ Skills

- ◇ Goals

- ◇ Remove:

- ◇ Impossible attackers

- ◇ Unrealistic assumptions

- ◇ Overly extreme scenarios

This is the **key filtering step**.

Step 4: Summarize & Structure (Often Using STRIDE)

- ◇ Organize remaining threats:

- ◇ Spoofing

- ◇ Tampering

- ◇ Repudiation

- ◇ Information disclosure

- ◇ Denial of service

- ◇ Elevation of privilege

- ◇ This converts:

- ◇ Creative ideas → structured threat list

The hTMM Process

Step 5: Move to Risk Assessment

- ◆ Now apply:
 - ◆ Likelihood
 - ◆ Impact
 - ◆ Risk prioritization
- ◆ This connects to:
 - ◆ Risk Assessment → Risk Management

The hTMM Process

Step 5: Move to Risk Assessment

How hTMM Maps:

	Four Questions	hTMM Step
◇ Now apply:		
◇ Likelihood	What are we building?	Step 1 (System identification)
◇ Impact	What can go wrong?	Step 2 + Step 3 + Step 4
◇ Risk prioritization	What should we do?	Step 5 (risk assessment → mitigation)
◇ This connects to:	How did we do?	✗ Missing
◇ Risk Assessment → Risk Management		

hTMM combines **creative**, **attacker-centric**, and **structured** methods to produce **realistic and prioritized threat scenarios**, but requires additional validation to ensure completeness (e.g., Security Cards → Random & Creative (Not Complete)).

Attack Trees

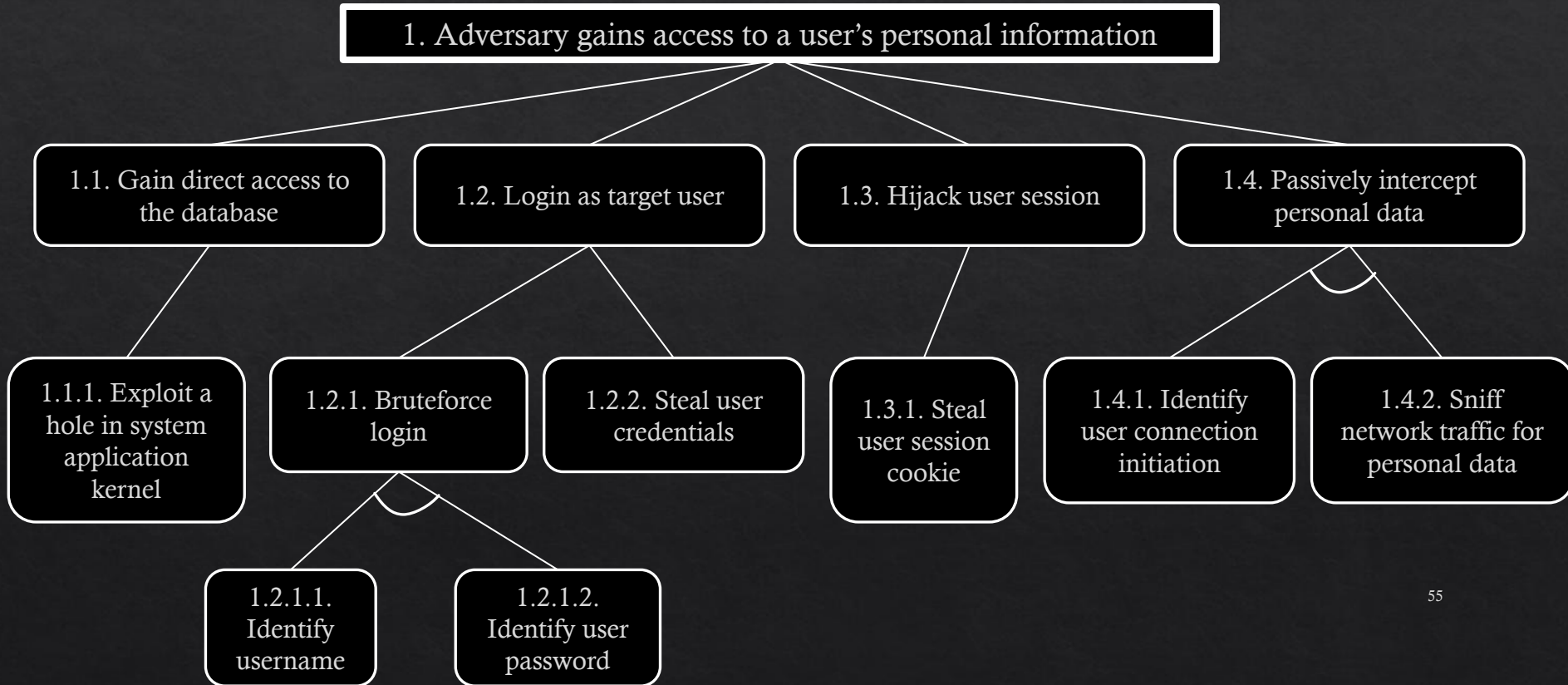
- ◇ Let us take a look at an Attack Tree (AT)
- ◇ ATs can
 - ◇ Describe security of systems and subsystems
 - ◇ Specify and decompose attack goals and steps to achieve it
 - ◇ Evaluate the severity of different vulnerabilities in the system
 - ◇ Helps making decision to improve security
- ◇ Often, you'll have multiple attack trees for a single system because the trees start with the attack goal and then build the attack path from there. Depending on a system's complexity, you might find potential attackers **have multiple goals**, requiring a new tree for each.

Attack Trees

◇ Structure

- ◇ Represent the attacks and countermeasures as a tree structure
- ◇ Root node is the goal of the attack
- ◇ Leaf nodes represents (atomic) attacks
- ◇ Uses logical gates to connect different attack events
 - ◇ E.g., uses of AND or OR gates
 - ◇ Attacker needs a swipe card AND pin number
 - ◇ Attacker needs to pick the lock OR break a window

Attack Trees



Attack Trees

1. Adversary gains access to a user's personal information

OR 1.1 Gain direct access to the database

1.1.1 Exploit a hole in system application or kernel

1.2 Log in as target user

OR 1.2.1 Brute-force login

AND 1.2.1.1 Identify username

1.2.1.2 Identify user password

1.2.2 Steal user credentials

1.3 Hijack user session

1.3.1 Steal user session cookie

1.4 Passively intercept personal data

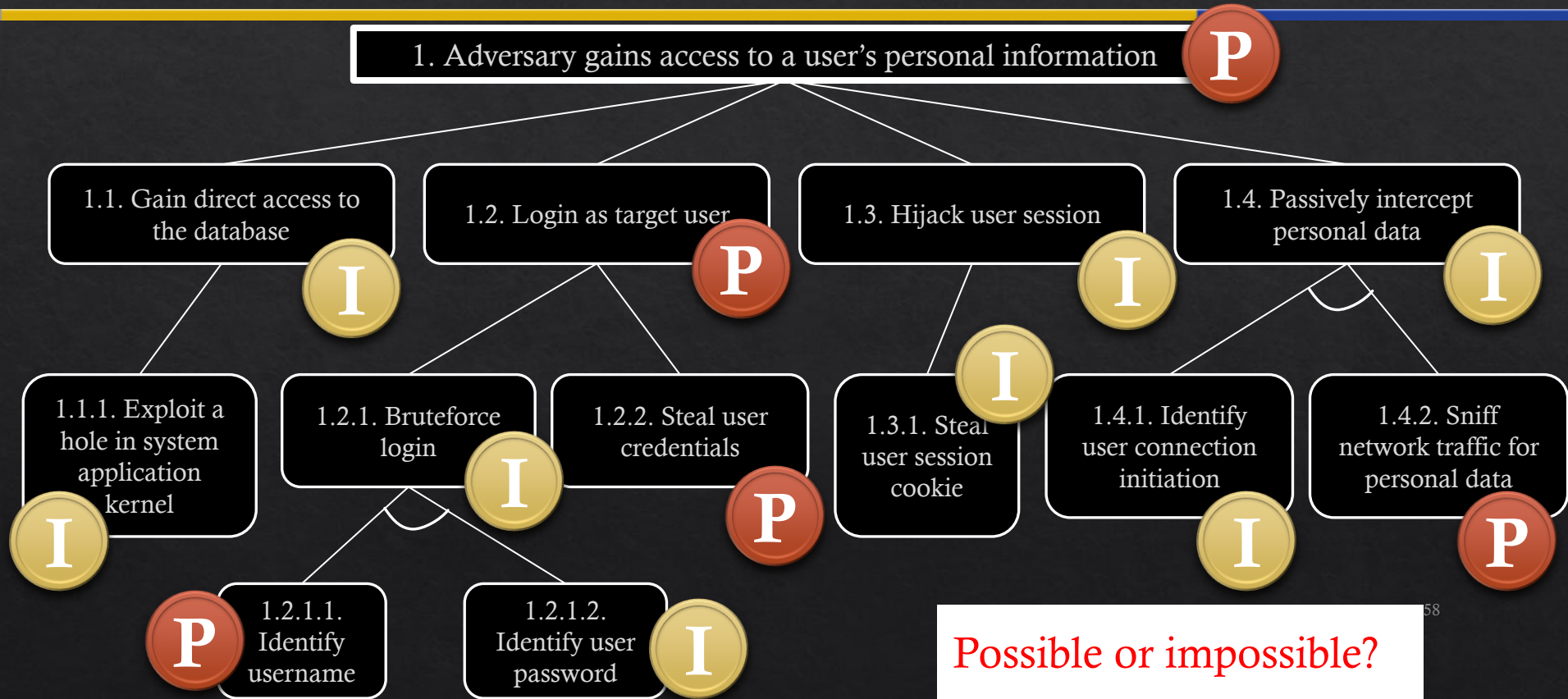
AND 1.4.1 Identify user connection initiation

1.4.2 Sniff network traffic for personal data

Attack Trees

- ◆ Leaf nodes can be assigned values to evaluate ATs
 - ◆ Boolean [e.g., possible/impossible]
 - ◆ Continuous [e.g., cost in \$, time]
 - ◆ Other [e.g., probability, impact, risk]
 - ◆ Combined [e.g., cheapest attack with high probability]

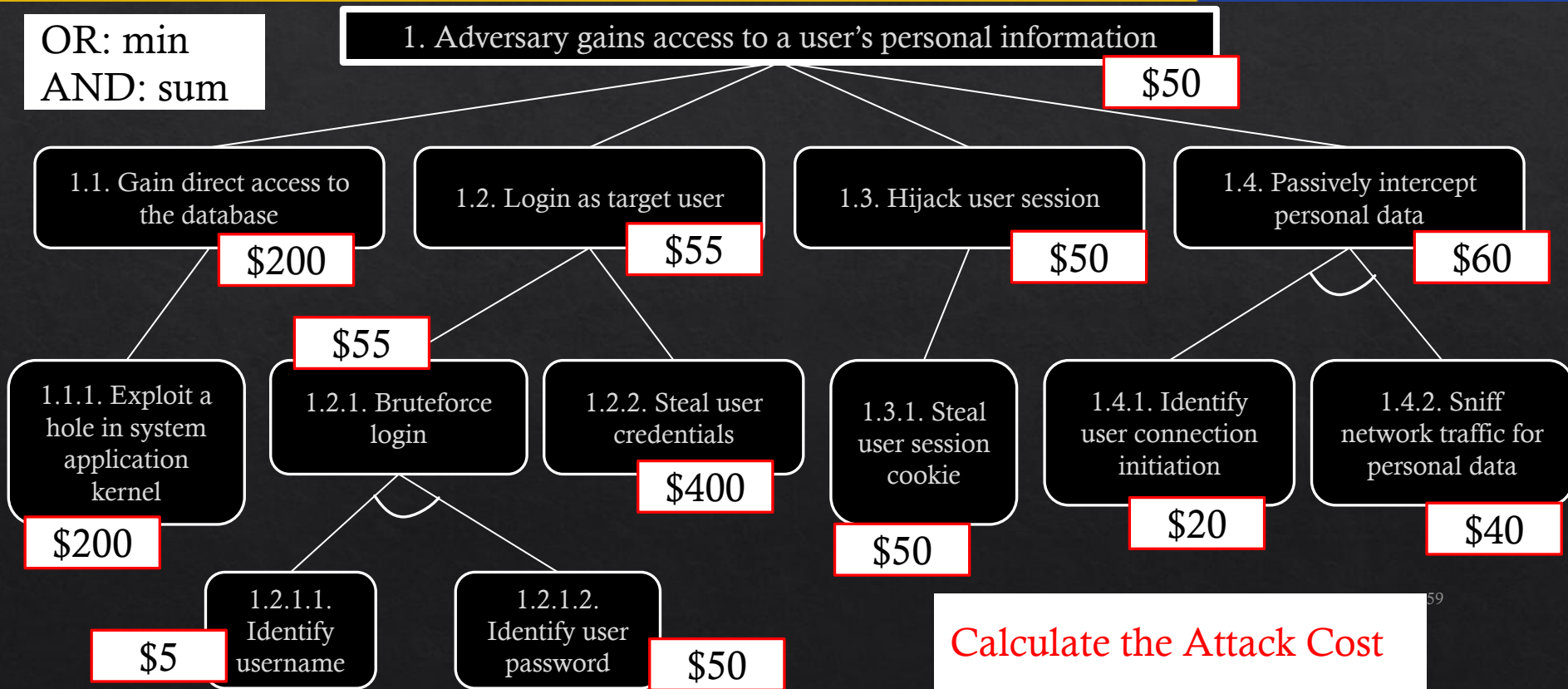
Attack Trees



Possible or impossible?

Attack Trees

OR: min
AND: sum



NIST 800-154

- ◇ It is data-centric threat modeling, focusing on protecting particular types of data within systems.
- ◇ It outlines four primary steps:
 - ◇ Identify and characterize the system and data of interest
 - ◇ Identify and select the attack vectors to be included in the model
 - ◇ Characterize the security controls for mitigating the attack vectors
 - ◇ Analyze the threat model
- ◇ **Never formally published**, the draft was published in March 2016

<https://csrc.nist.gov/pubs/sp/800/154/ipd>

Who can use threat modelling?

- ◇ Software Development Teams and Testers
 - ◇ Threat modeling helps them find and address design issues early in the development process so that they can build security directly into the Software Development Life Cycle (SDLC)
- ◇ Systems Architecture and Operations Teams
 - ◇ Threat modeling helps people who bring together software components to address threats when they: Design systems, Select components, Prepare for deployment. With threat modelling, they can find security and privacy threats so that they can make informed tradeoffs that address how to handle threats.
- ◇ Project Managers
- ◇ Penetration Testers
- ◇ Security Teams

Summary

- ◊ Threat Modelling Vs Risk Assessment
- ◊ Four key questions
- ◊ Threat Modelling Methods and Techniques
- ◊ Threat Modelling Users

Additional Materials

- ◇ Threat Modeling Frameworks: The Unsung Heroes of Cybersecurity
 - ◇ <https://pantheon.io/learning-center/threat-modeling-frameworks>
- ◇ What is Threat Modeling and Why Is It Important?
 - ◇ https://www.youtube.com/watch?v=h_BC6OMWDbA
- ◇ OWASP Threat Modelling Project
 - ◇ <https://owasp.org/www-project-threat-modeling/>
- ◇ Threat Modelling Process (This provides a detailed example and the usage of STRIDE)
 - ◇ https://owasp.org/www-community/Threat_Modeling_Process
- ◇ Threat Modelling vs Risk Assessment: Understanding the Difference
 - ◇ https://www.practical-devsecops.com/threat-modeling-vs-risk-assessment/?srsltid=AfmBOopDmr5uNMUx_M0CC-xoItt0r8i1-Bb66--GJBGBE60mwQLm7GOx